

Technical Appendix

Is Imagination Derived from Hallucination?

A Cross-Taxonomy Evaluation of Imagination and Hallucination in Large Language Models

Anonymous submission

This document is the technical appendix for the submission *Is Imagination Derived from Hallucination? A Cross-Taxonomy Evaluation of Imagination and Hallucination in Large Language Models*. It collects the extended material referenced from the main paper. Sections A–E document the instrument, Sections F–J the supporting analyses, Sections K–N what was run and on what, and Sections O–R the limits of the design. The main paper is self-contained; nothing here is required to follow its argument. Section letters, figure numbers (Figure B1, . . .) and table numbers (Table D1) below are local to this appendix; references to the main paper are given as explicit numbers.

A Full Taxonomy and Facets

The full facet tables for all seven imagination and ten hallucination subtypes, with white-box-signal lists, will be reproduced in the camera-ready and are available in the released taxonomy file. Each subtype has between three and five facets; each facet has at least one white-box-signal entry.

B Atom Audit Matrix

The audit matrix $M \in \{-1, 0, +1\}^{116 \times 17}$ records the signed contribution of each raw atom entry to each of the 17 subtype scores; 103 of the 116 atoms belong to the eight imagination-bearing families, the remaining 13 to the hallucination-calibration family and the zero-weight consistency probe. Figure B1 visualizes the sparsity pattern and the separation between I-side support atoms and H-side penalty atoms. Per-cell shared atom sets $S_{a,b}$ for the correlation table are released alongside the matrix.

C Cross-Axis Scoring Design

The scoring design keeps imagination-side support signals separate from hallucination-side penalty signals when a task family uses related evidence. In PropConj, the grounding signal for I_{assoc} is distinct from the context-support check for H_{ctx} . GCW likewise separates narrative support from unsupported-detail evidence, while CJST separates counterfactual compliance from logic violations. For the cells

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

reported in Table 5 of the main paper, these definitions leave $S_{a,b} = \emptyset$, and the atom-partial correlations are therefore nearly identical to the unadjusted correlations. The partial analysis consequently tests overlap in the scoring formulas used for the reported results.

D Decisive Cells: Full Table

The main paper shows only the strongest decisive cells. Table D1 lists every cell that clears BH-FDR at $q \leq 0.05$ in its own view and in both partial controls, grouped by view, with the raw Spearman correlation, its atom- and capability-partial versions, the three q values, and the bootstrap interval. Twenty-three cells qualify: six in the raw view, ten in the gated view, and seven in the residual view, of which twenty-two are negative. The negative cells recur across views rather than appearing once: five distinct pairs qualify in all three views (narrative $\times$ detail, narrative $\times$ drift, counterfactual $\times$ logic, code $\times$ fact, hypothesis $\times$ boundary) and three more in two views. The one positive cell qualifies in a single view, at the largest corrected q in the table.

E Forest Plot of Decisive Cells

Figure E1 puts effect sizes and uncertainty next to the counts above. Each decisive cell is drawn three times, once per estimator (raw, atom-partial, capability-partial), with 95% bootstrap intervals. Twenty-two intervals lie entirely below zero and one, analogy $\times$ detail in the gated view, lies entirely above it. For almost every cell the three estimators fall within a few hundredths of each other, so the couplings do not depend on which control is applied, and the widest intervals belong to the smallest effects, not to the strongest ones.

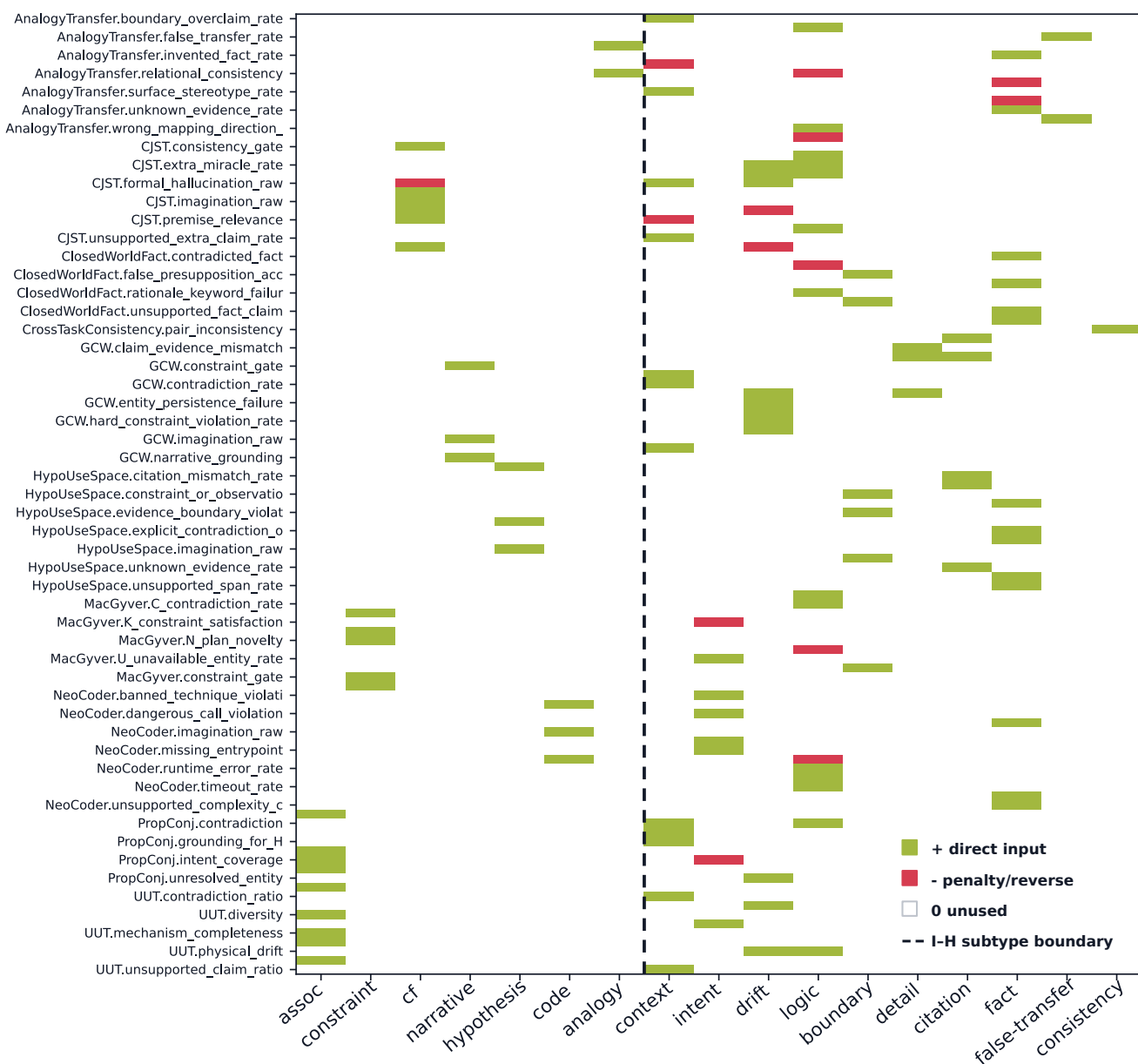


Figure B1: Signed atom-to-subtype dependency matrix. Green: positive contributions; red: penalty signals; white: unused. The block structure is the substrate for the shared-atom partial Spearman control.

View	Cell ($I_a \times H_b$)	n	ρ	ρ^a	ρ^c	q^v	q^a	q^c	95% CI
Raw	Narrative $\times$ Detail	79	-0.60	-0.60	-0.51	$< 10^{-6}$	$< 10^{-6}$	1.5×10^{-4}	[-0.75, -0.41]
Raw	CF $\times$ Logic	79	-0.56	-0.56	-0.39	1.8×10^{-6}	1.8×10^{-6}	0.006	[-0.71, -0.36]
Raw	Hypothesis $\times$ Boundary	79	-0.56	-0.56	-0.45	1.8×10^{-6}	1.8×10^{-6}	0.001	[-0.70, -0.38]
Raw	Code $\times$ Fact	79	-0.52	-0.52	-0.40	1.4×10^{-5}	1.4×10^{-5}	0.005	[-0.69, -0.31]
Raw	Narrative $\times$ Drift	79	-0.52	-0.52	-0.36	1.4×10^{-5}	1.4×10^{-5}	0.013	[-0.70, -0.30]
Raw	Narrative $\times$ Context	79	-0.51	-0.51	-0.32	1.4×10^{-5}	1.4×10^{-5}	0.044	[-0.67, -0.31]
Gated	Analogy $\times$ False-Transfer	79	-0.72	-0.72	-0.74	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	[-0.83, -0.58]
Gated	Narrative $\times$ Detail	79	-0.66	-0.66	-0.58	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	[-0.78, -0.49]
Gated	Narrative $\times$ Drift	79	-0.63	-0.63	-0.51	$< 10^{-6}$	$< 10^{-6}$	4.4×10^{-5}	[-0.78, -0.43]
Gated	CF $\times$ Logic	79	-0.59	-0.59	-0.40	$< 10^{-6}$	$< 10^{-6}$	0.002	[-0.73, -0.40]
Gated	Hypothesis $\times$ Boundary	79	-0.55	-0.55	-0.43	2.4×10^{-6}	2.4×10^{-6}	0.002	[-0.70, -0.36]
Gated	Narrative $\times$ Context	79	-0.53	-0.53	-0.33	5.3×10^{-6}	5.3×10^{-6}	0.022	[-0.69, -0.32]
Gated	Code $\times$ Fact	79	-0.52	-0.52	-0.41	6.2×10^{-6}	6.2×10^{-6}	0.002	[-0.69, -0.32]
Gated	Hypothesis $\times$ Citation	79	-0.51	-0.51	-0.41	1.2×10^{-5}	1.2×10^{-5}	0.002	[-0.67, -0.31]
Gated	Analogy $\times$ Fact	79	-0.36	-0.36	-0.36	0.004	0.005	0.010	[-0.57, -0.13]
Gated	Analogy $\times$ Detail	79	0.27	0.27	0.33	0.039	0.042	0.019	[0.03, 0.48]
Residual	Analogy $\times$ False-Transfer	79	-0.81	-0.81	-0.81	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	[-0.88, -0.69]
Residual	Hypothesis $\times$ Citation	79	-0.69	-0.69	-0.65	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	[-0.79, -0.55]
Residual	CF $\times$ Logic	79	-0.61	-0.61	-0.49	$< 10^{-6}$	$< 10^{-6}$	6.6×10^{-5}	[-0.75, -0.42]
Residual	Code $\times$ Fact	79	-0.60	-0.60	-0.44	$< 10^{-6}$	$< 10^{-6}$	5.0×10^{-4}	[-0.74, -0.42]
Residual	Narrative $\times$ Detail	79	-0.56	-0.56	-0.51	$< 10^{-6}$	$< 10^{-6}$	3.6×10^{-5}	[-0.71, -0.37]
Residual	Narrative $\times$ Drift	79	-0.52	-0.52	-0.45	8.5×10^{-6}	8.5×10^{-6}	3.6×10^{-4}	[-0.70, -0.32]
Residual	Hypothesis $\times$ Boundary	79	-0.48	-0.48	-0.38	7.4×10^{-5}	8.4×10^{-5}	0.006	[-0.64, -0.27]

Table D1: All 23 triple-FDR-significant cells in the main panel; column notation follows Table 5 of the main paper.

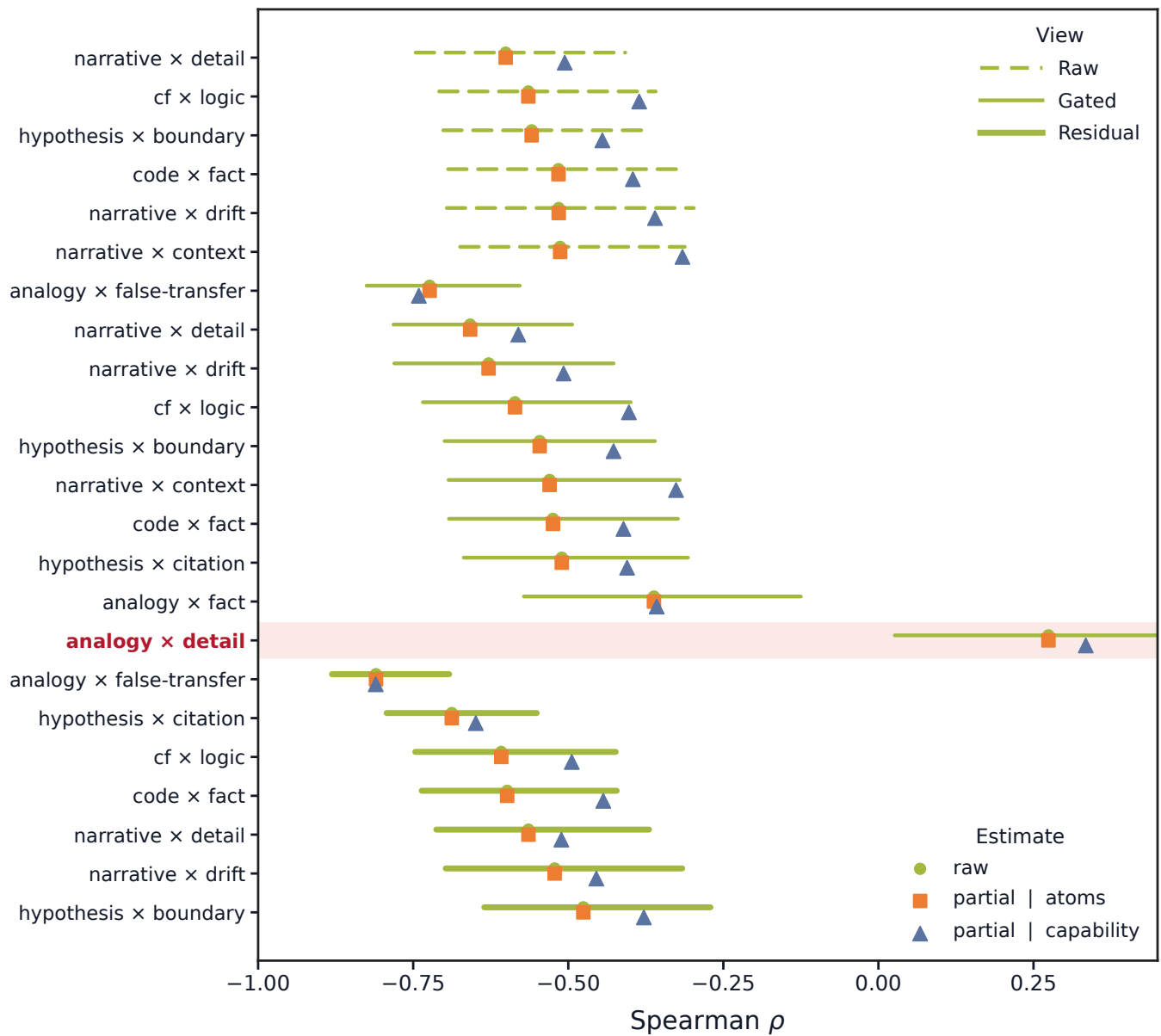


Figure E1: Forest plot of the 23 decisive cells. Raw, atom-partial, and capability-partial estimates with 95% bootstrap CIs.

F Paired Discriminative Power and Reliability

We first ask how much detection power the shared items buy. On the 58 models with complete coverage, we tested all 1,653 model pairs with a Wilcoxon signed-rank test over shared items and a Mann–Whitney test over the same scores, followed by BH-FDR correction. The paired analysis separates 54 pairs, whereas the unpaired analysis separates none (Figure F1). Holding the item fixed removes prompt difficulty from the model contrast, which accounts for the gain.

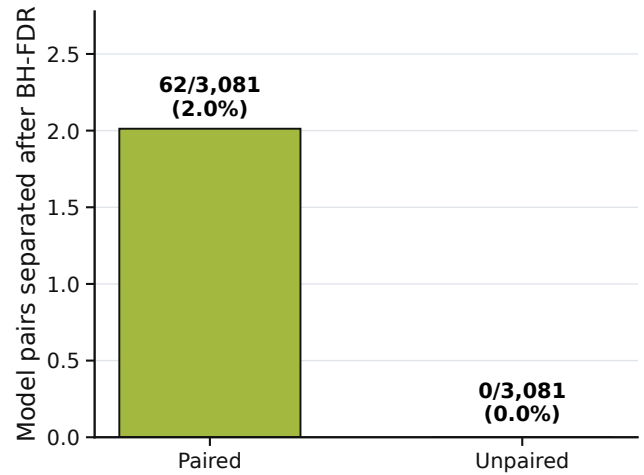


Figure F1: Model pairs distinguishable after BH-FDR under paired and unpaired analyses of the same scores.

How many items a comparison needs is a separate question. We decomposed model, item, and interaction variance for the same 58 complete models and projected relative reliability G and absolute reliability Φ from 10 to 160 items. At 80 items, imagination reaches $G=0.567$ and $\Phi=0.248$, while hallucination reaches $G=0.765$ and $\Phi=0.663$ (Figure F2). The lower absolute reliability of imagination reflects the larger share of variance carried by item difficulty. Panel-level relative comparisons are therefore on firmer ground than absolute score transfer to another prompt set.

G Sensitivity Analysis

We deleted each of the 18 providers in turn and applied the same fixed scoring rule to 43 models generated at temperatures 0.85 and 0, retaining the same 73–80 valid items per model across settings. Decisive-cell retention is 17–100% in the raw view, 70–100% in the gated view, and 86–100% in the residual view; the aggregate correlation changes from -0.332 to -0.722 at temperature 0, with cross-temperature rank correlations of 0.585 for imagination and 0.877 for hallucination (Figures G1 and G2). The raw view is sensitive to removal of a large provider, whereas the gated and residual views are more stable and deterministic decoding does not remove the negative association. The gated and residual findings thus hold across provider composition and both decoding settings; the raw view does not carry a claim on its own.

H Subtype Profiles by Tier

The main paper reports a null tier difference on both axes; this section shows the profiles behind it. Figure H1 plots the mean score of each imagination and destructive-hallucination subtype for the 21 frontier and the 58 comparison models. The two profiles overlap on both axes, with no subtype on which one tier is consistently outside the other. The tiers differ in product position and release recency, not in the subtype composition of either axis. The figure is included for

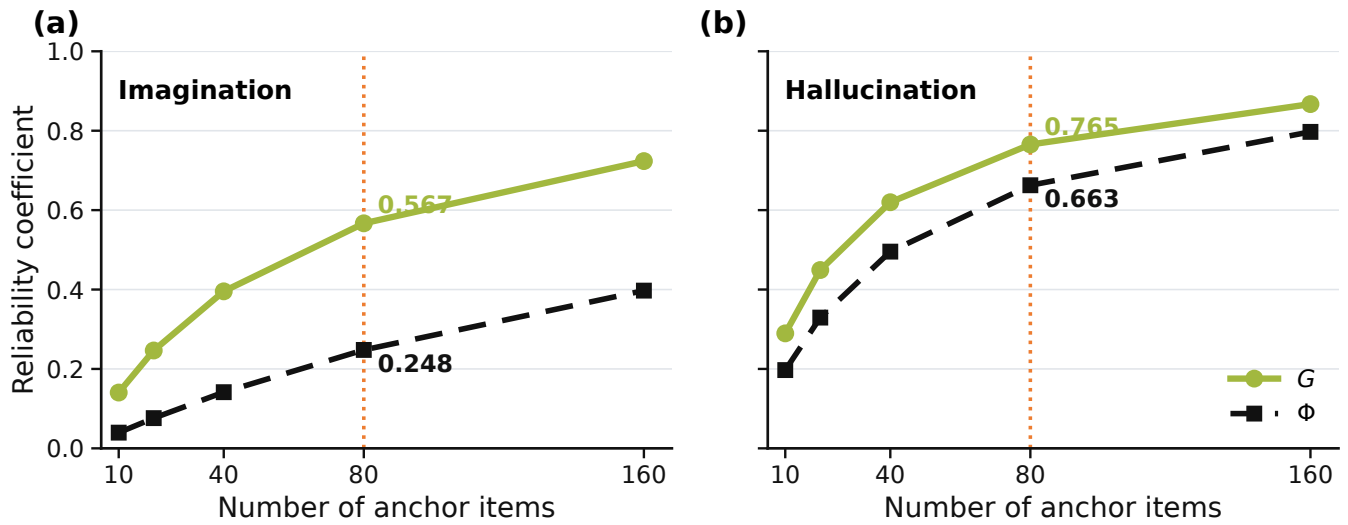


Figure F2: Projected reliability as the number of anchor items changes. Panel (a) reports imagination and panel (b) hallucination; solid green lines show relative reliability G , and dashed black lines show absolute reliability Φ .

orientation only; every inferential claim rests on the triple-FDR map (Figure 2 of the main paper) and on the tests in section “Who Trades Imagination for Reliability?”.

I Sensitivity Across Scoring Views

Some cells read differently under the raw, gated, and residual scoring views. The released tables report all three estimates and flag cells whose largest difference is $|\Delta\rho| \geq 0.30$. A flagged cell is not removed automatically, because retention still requires the view test and both partial controls to pass the same false-discovery threshold. Large cross-view differences instead indicate that a support gate or correction term contributes to the observed association. Reporting the three estimates prevents a single-view result from being read as view-invariant.

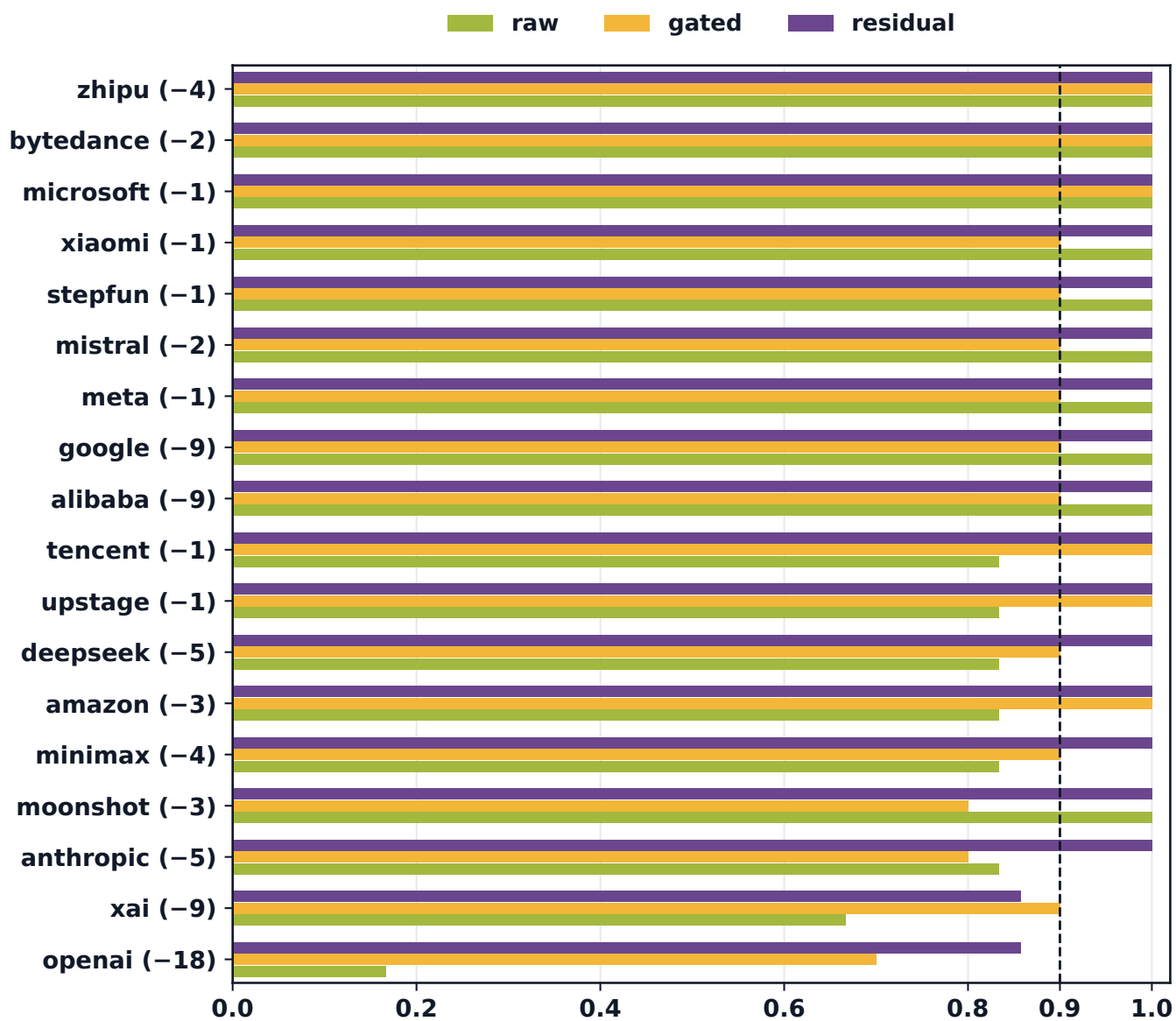


Figure G1: Provider leave-one-out retention of decisive cells, by analysis view.

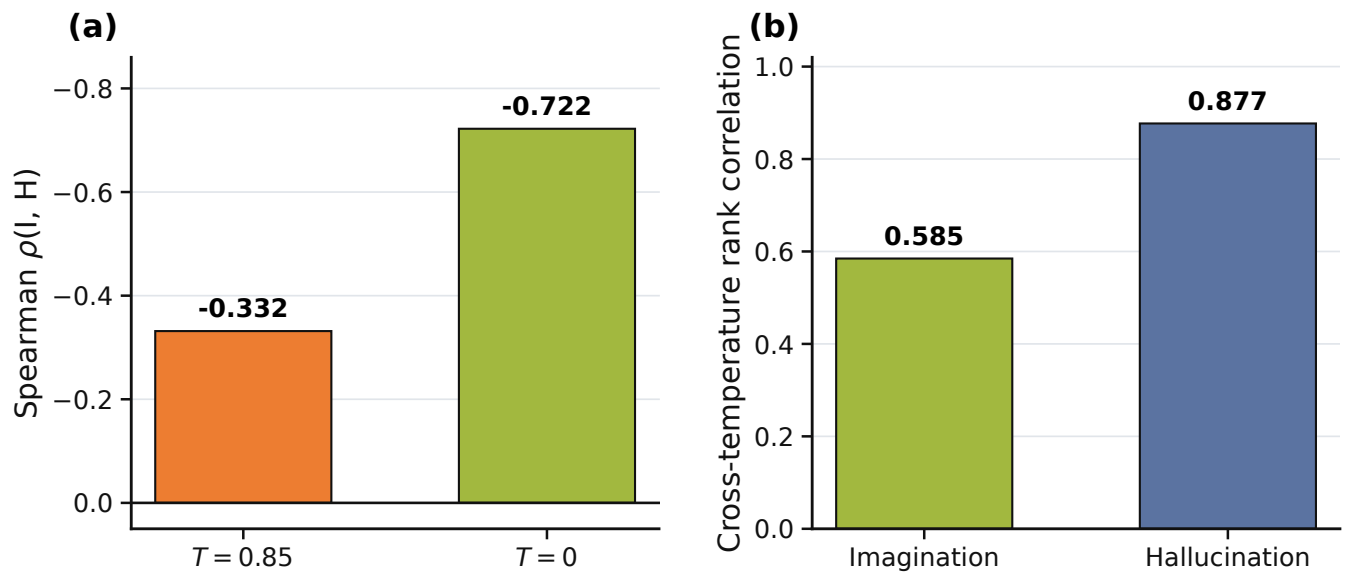


Figure G2: Sensitivity to decoding temperature on 43 matched models. Panel (a) compares the aggregate imagination–hallucination correlation, and panel (b) reports rank agreement between the two temperature settings on each axis.

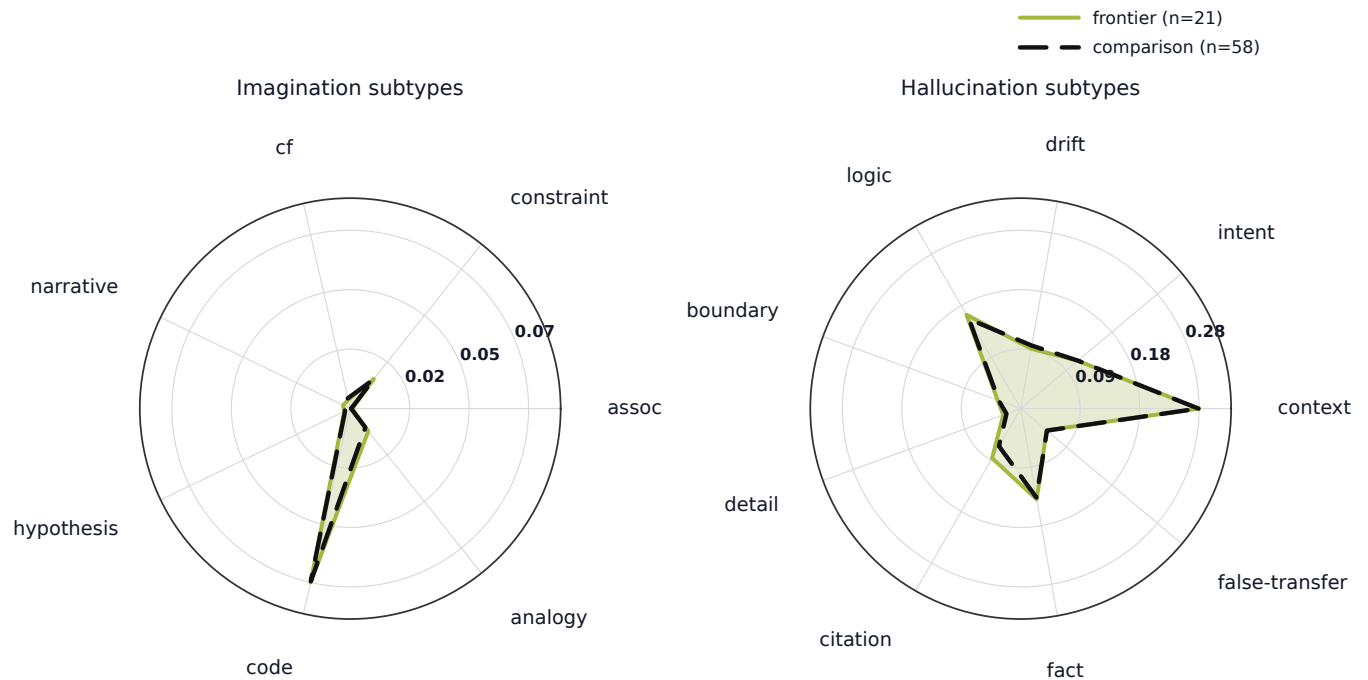


Figure H1: Subtype profiles by model tier (frontier vs comparison).

J Case Study: Six Audited Outputs

Six outputs make the joint reading concrete: for each of three task families we pair one output whose invention stays inside the support boundary with one whose invention crosses it. The examples are qualitative illustrations from the benchmark corpus and carry no statistical claim. The three families carry different imagination subtypes; narrative $\times$ detail and cf $\times$ logic are decisive cells of the correlation map, while the code pair is illustrative because code $\times$ intent does not survive the triple-FDR criterion. Table J1 states what each prompt provides and licenses; Tables J2, J3, and J4 present the three pairs.

Grounded Fiction (GCW). On the card *The Broken Compass*, one output turns the floor itself into the way out: Priya

Family	The Prompt Fixes	Invention Licensed	Paired H Reading
GCW (I_{narr})	Closed fact sheet (F1–F6), hard-constraint sheet (C1–C3), allowed-entity list	Sensory detail, dialogue, small actions	Unsupported detail, fact conflict (H_{det} , H_{fact})
CJST (I_{cf})	One impossible premise; all other physics, biology, and institutions stay normal; protected variables	Twelve consequences (immediate, adaptive, second-order), each with a causal chain from the premise	Extra magic, impossible physics (H_{logic})
NeoCoder (I_{code})	Entry-point signature, allowed-imports list, denial state banning standard techniques, JSON-only output contract	Alternative algorithmic strategy under the denial	Contract or constraint miss, fabricated API (H_{int} , H_{fact})

Table J1: Task settings for the three case-study families: what each prompt fixes, what it licenses, and which hallucination subtypes read the same output.

	<i>The Broken Compass</i>	<i>The Fog Market</i>
Fact Sheet (Excerpt)	Priya remembers left and right by touch; the marble floor has raised cracks; visitors must exit without panic	Otto identifies coins by weight; fog hides the stall signs; a lost coin pouch must return to its owner
Hard Constraints	Compass stays broken; lights keep flickering; no emergency crew enters	Fog does not clear; Otto cannot see through it; no town guard solves it
Model’s Move	Reads the raised cracks by touch and uses them as an exit map	Asserts that coin weight identifies Sima’s stall and that Sima knows the owner
Support Status	Every element used is on the sheet	Both load-bearing claims are unsupported
Audit Reading	Top-scale I_{narr} ; no unsupported-detail or fact evidence	H_{det} , H_{fact} ; broken output contract (H_{int})

Table J2: The two GCW outputs of the case study at a glance. The same license to invent splits into grounded invention and fabricated shortcut exactly at the support boundary.

reads the raised cracks in the marble with her touch-based left–right memory and uses them as an exit map. The turn appears nowhere in the fact sheet, yet every element it uses does, and WHITEBOARD scored it at the top of the narrative scale with no unsupported-detail evidence. On *The Fog Market*, a second output reaches the goal only by asserting what the sheet never supports: Otto “recognizes the weight of the coins as those used at Sima’s spice stall,” and Sima “confirms the pouch belongs to a regular customer who always buys cumin.” Coin weight identifies denominations, not a stall’s customers, and no fact grants Sima knowledge of the owner; WHITEBOARD flags both claims (H_{det} , H_{fact}).

Counterfactual Extension (CJST). Both outputs were rated at the top of the counterfactual scale; only one keeps the premise closed. Under the premise “every cup remembers the last drink poured into it,” the first output derives consequences that never leave the premise: people stop sharing cups, baristas check a cup’s memory before refilling it, parents discard cups after allergenic drinks; each consequence carries a causal

	Cup Memory	Glowing Footsteps
Premise	Every cup remembers the last drink poured into it	Footsteps leave faint glowing marks for one minute
Hard Constraints	Only this fact is impossible; physics, biology, institutions stay normal	Same closure; no extra magic beyond the premise
Model’s Move	Shared cups become unappealing; baristas check cup memories; risky cups are discarded	Flooring products tune glow visibility; security systems detect the glow
Support Status	Every consequence is carried by the premise	The glow is treated as an engineerable substrate the premise never licenses
Audit Reading	Top-scale I_{cf} ; no logic-violation evidence	Top-scale I_{cf} , but top-scale H_{logic} as well

Table J3: The two CJST outputs. Both are maximally imaginative on the imagination axis; they differ only in whether the invention stays inside the counterfactual premise.

	Clean Delivery	Broken Contract
Task	Count 4-connected islands in a 0/1 grid	Same problem, same contract
Contract	One JSON object, required keys, no text outside it, empty imports, fixed signature	Identical
Model’s Move	Recursive depth-first search with a visited matrix, returned as a single parseable object	Reasoning text emitted before the JSON object
Support Status	Contract respected end to end	Output unparseable; the code never reaches the hidden tests
Audit Reading	Top-scale I_{code} ; no H evidence on any subtype	Top-scale algorithmic content, but a contract miss (H_{int})

Table J4: The two NeoCoder outputs on the same count-islands task. The pair isolates the delivery contract: identical problem, identical license, opposite H_{int} readings.

chain anchored in the premise, and WHITEBOARD records no impossible-physics evidence. Under the premise “footsteps leave faint glowing marks for one minute,” the second output starts to engineer the miracle: flooring products that “minimize or enhance glow visibility” and security systems with “one-minute glow detection algorithms.” The premise licenses the glow; nothing licenses machines that read or amplify it. The prompt names this move explicitly, no extra magic beyond the premise, and WHITEBOARD scored it at the top of the logic-violation scale (H_{logic}).

Creative Code (NeoCoder). The third pair shares one problem, counting 4-connected islands in a 0/1 grid, under the same JSON-only contract: return exactly one object with the required keys, no markdown, no text before or after it. The first output delivers a recursive depth-first search behind the required entry-point signature, respects the empty-imports list, and parses cleanly; WHITEBOARD records no hallucination evidence on any subtype. The second opens with reasoning text before the JSON object, so the object never parses and the code inside never reaches the hidden tests. WHITEBOARD still rated its algorithmic content at the top of the code scale, but the delivery is exactly what H_{int} records: a contract miss that no amount of algorithmic creativity repairs.

A distance-only creativity score cannot tell the outputs in any of these pairs apart: within each pair both leave the obvious answer, and in every pair both outputs were rated at the top of the imagination scale by the scorer. The joint reading separates them, and for two of the three families the separation it makes on single outputs is the one the panel-scale couplings recover across 79 models: narrative $\times$ detail at $\rho = -0.66$ and $cf \times logic$ at -0.61 (Table 5 of the main paper).

K Panel Manifest

The full panel manifest (model identifiers, providers, families, release dates, tier assignment, and per-item coverage) is released with the artifact. Provider distribution of the 79-model panel: OpenAI (18), xAI (9), Google (9), Alibaba (9), DeepSeek (5), Anthropic (5), MiniMax (4), Zhipu (4), Moonshot (3), Amazon (3), ByteDance (2), Mistral (2), and one model each from StepFun, Meta, Tencent, Microsoft, Upstage, and Xiaomi. The frontier tier covers the GPT-4.1 and GPT-5 families (OpenAI 2026), Claude Opus 4.6/4.7 and Sonnet 4/4.5/4.6 (Anthropic 2026), Gemini 3.1 Pro (Google DeepMind 2026), and Grok 4.x (xAI 2026); the comparison tier adds DeepSeek (DeepSeek 2026), Qwen (Qwen Team 2026a,b), Kimi (Moonshot AI 2026a,b), MiMo (Xiaomi MiMo 2026), Gemma (Google DeepMind 2026), GLM (GLM Team 2025; Z.ai 2026), Hunyuan (Tencent Hy Team 2026), Llama (Meta 2024), MiniMax (MiniMax 2026a,b), and earlier GPT, Claude, Gemini, and ByteDance Seed variants (ByteDance Seed 2025, 2026). Tier assignment reads model metadata only and never reads scores: 21 models are frontier and 58 comparison. The panel is fully crossed by construction; the 91 model-item cells that returned no scorable output (79 with no generation, 12 failing the task’s parse contract) are recorded as missing rather than refilled, which is why realized item overlap is 0.98 and not 1.00, and why the complete-case analyses (paired power, variance components) run on the 58 models that carry a scored output for every one of the 80 items.

L Scoring Configuration Reproducibility

Every downstream analysis should run off one scoring configuration. We compared the configuration identifier attached to each of the 6,443 matched annotation rows and rebuilt model aggregates from their per-task contributions. Every matched row uses the same configuration, and the maximum reconstruction errors are 1.03×10^{-15} for imagination and 4.19×10^{-15} for hallucination. Model identity, tier, provider, and release date are not inputs to runtime scoring, so the reconstruction does not depend on model-specific settings.

Resource Versions and Access. Embedding atoms use the `all-mpnet-base-v2` encoder at revision `e8c3b32` (Reimers and Gurevych 2019; Sentence Transformers 2025) under the model repository’s Apache 2.0 license. Lexical resources are SWOW-EN2018 R123 (De Deyne et al. 2019), Word Norms 2 (Buchanan, Valentine, and Maxwell 2019), and WordNet 3.0 (Miller 1995). The raw SWOW file is obtained from the official project under its research-use terms and is not redistributed. Word Norms 2 data and its GPL-3.0

reference code are public, and the WordNet 3.0 license permits use, copying, modification, and distribution with its notices retained. The official Arena reference is the LM Arena method and leaderboard dataset (Chiang et al. 2024; LM Arena 2026), using the 2026-04-27 snapshot. The frozen configuration records the final-panel Seed 1.6 anchor as a value from the third-party BenchLM model page (BenchLM 2026), and not an official LMArena row. The current page exposes no sourced score, so the citation documents that limitation; it does not validate the value. The common-answer banks are author-built: the static bank and curated overlay are included with the artifact; mined extensions require regeneration from the documented settings and access to the listed model APIs.

M Held-Out Annotator Agreement

Two annotators independently rated 6,640 outputs on 0–4 imagination and unsupported-content scales. The ratings of annotator C selected the subtype score boundaries; the ratings of annotator D were never read by any fitting step. Machine correlations with the held-out annotator are 0.710 and 0.632 at output level ($n=6,443$) and 0.546 and 0.634 at model level ($n=79$). Against the two raters’ mean, the corresponding values are 0.716 and 0.694, and 0.560 and 0.670. Reversing the split and selecting boundaries on annotator D gives model-level correlations of 0.511 and 0.644 on annotator C, against 0.546 and 0.634 in the deployed direction. The two boundary sets do not match cell by cell, but their model-score rankings agree at $\rho=0.977$ and 0.996 for imagination and hallucination. The raw view, which the boundary and residual calibration does not touch, agrees with the held-out annotator at $\rho=0.50$ and 0.57 at model level (Figure M1a). Inter-annotator correlations are 0.712 for imagination and 0.759 for unsupported content, with agreement within one point at 97.70% and 91.07%. Licensing rates span 18.74%–22.85% across subtypes and false transfer does not differ from the others (Fisher $p=.218$). The hold-out is at the rater level. The held-out annotator worked from the same rubric on the same outputs, so this design controls for rater-specific fitting, not for circularity introduced by the rubric itself.

N Prompt Examples

The crossed panel used 80 anchor prompts from eight scored components. We sent the same 80 task IDs to all 79 models. Each request contained a task-specific system message and one user message, and asked for a single model response. Figures N1 and N2 pair prompt passages with selected fields from the saved responses for all eight components.

N.1 Request Format

The system message defined the benchmark role and output format. The user message supplied the task instance and its support rule, followed by the required JSON schema and output count. Depending on the task, the instance could contain a tool inventory, a fact sheet, an evidence table, or a function signature. Responses had to contain JSON without surrounding prose. We did not add demonstrations or follow-up turns for particular models.

The support rule was therefore part of the input seen by the model. Its form varied by task. Some prompts limited proposals to physical properties or listed tools, while others licensed a single counterfactual premise, a fictional world, or an analogy. The prompt also stated which additions fell outside that license.

N.2 Task-Specific Instructions

UUT and PropConj. UUT requested eight unusual but physically implementable uses of an ordinary object and ruled out invented capabilities. PropConj requested six real objects that satisfied every named property. An uncommon answer was acceptable only when it remained possible and met the full conjunction.

MacGyver and HypoUseSpace. MacGyver supplied a goal, a tool inventory, and physical or safety constraints. Plans could combine listed tools but could not introduce new equipment. HypoUseSpace instead supplied entities, relations, predicates, and evidence IDs. Its answers had to cite the records used to support the proposed hypothesis.

CJST and GCW. CJST introduced one impossible premise and requested immediate, adaptive, and second-order consequences while keeping ordinary constraints in force. GCW paired a story card with a fact sheet and a list of forbidden claims. Narrative detail was open, but the stated world facts and exclusions remained binding.

NeoCoder and AnalogyTransfer. NeoCoder fixed the entry point, required behavior, permitted imports, and prohibited techniques for an executable coding task. AnalogyTransfer named a source system and a target domain, then specified which source relations could transfer and which source details could not be treated as literal facts about the target.

N.3 Source of the Examples

The cards draw from the prompt manifest and task results saved with one panel report. The build script checks each displayed prompt passage against the full prompt and each response field value against the raw model output; it stops if either check fails. A bracketed ellipsis marks text omitted between prompt passages. Each task ID locates the complete prompt and response. Colors distinguish components and do not encode scores.

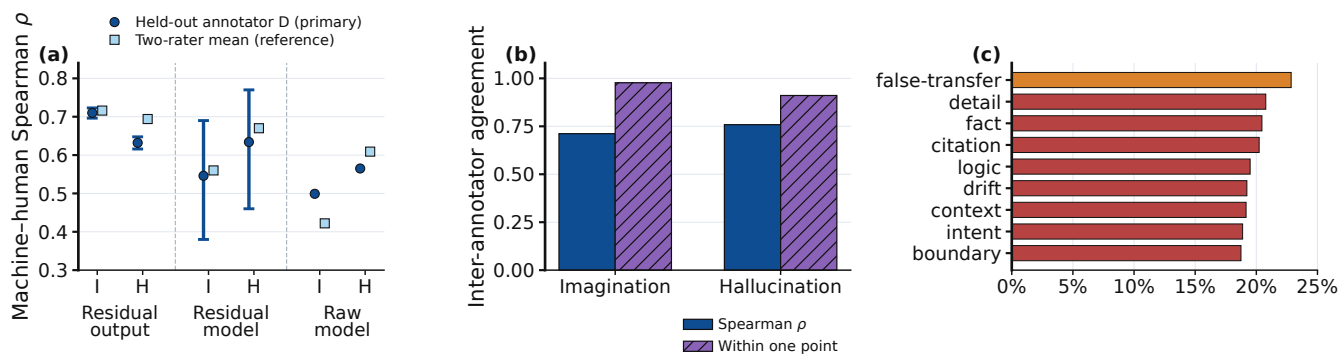


Figure M1: Human-rating agreement. Panel (a) compares the residual scores with the held-out annotator and the two-rater mean at output and model levels; the raw model-level points provide an uncalibrated reference. Filled dark circles mark the held-out comparison, and light squares mark the two-rater mean. Error bars are 95% bootstrap intervals for the held-out comparisons. Panel (b) reports inter-annotator agreement, and panel (c) shows the licensed share of unsupported content by subtype.

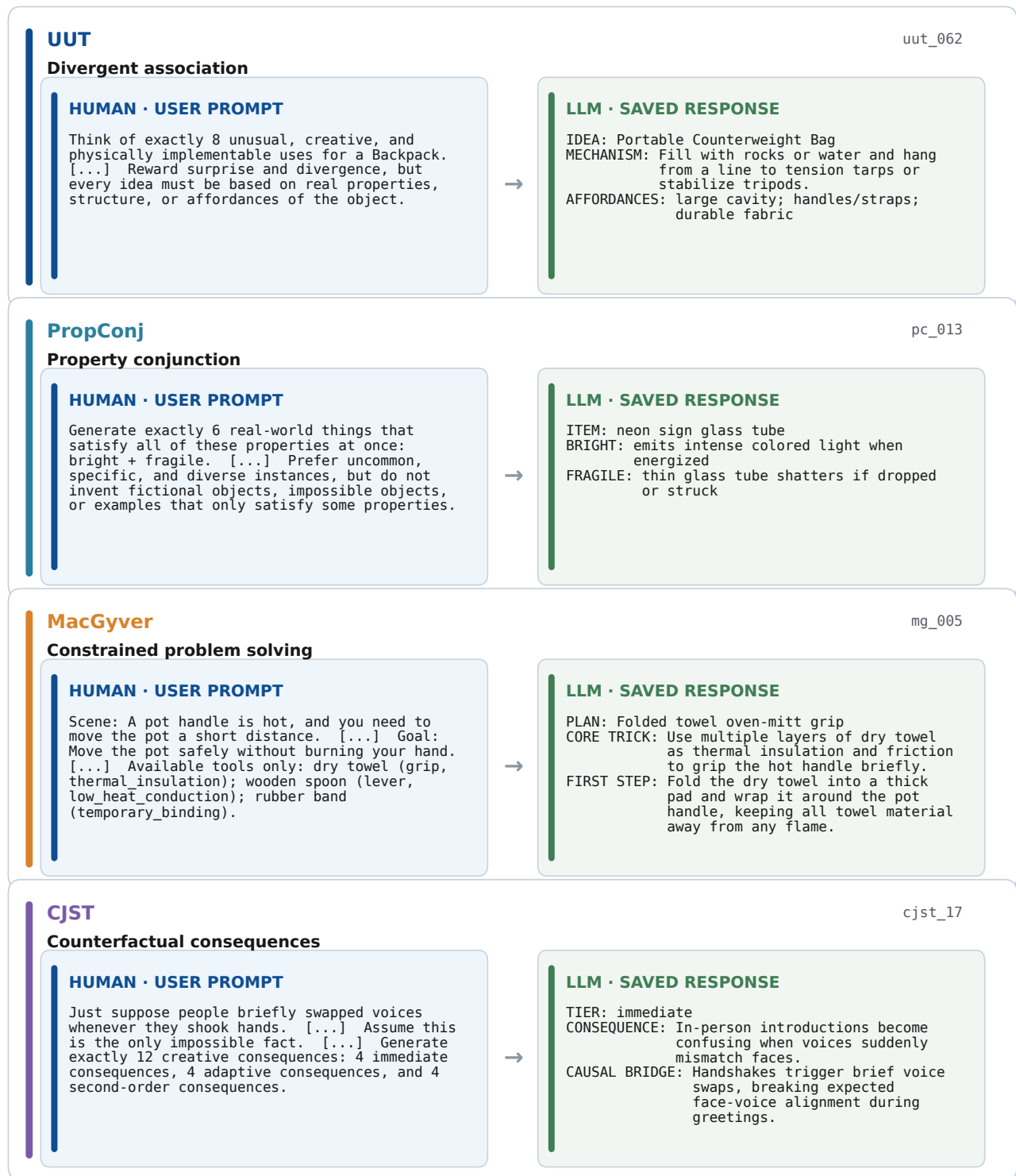


Figure N1: Human-LLM examples for UUT, PropConj, MacGyver, and CJST. Each row places evaluated user-prompt passages on the left and selected field values from the corresponding saved response on the right. Bracketed ellipses mark omitted prompt text, and task IDs identify the full saved records. Colors distinguish benchmark components and do not encode scores.

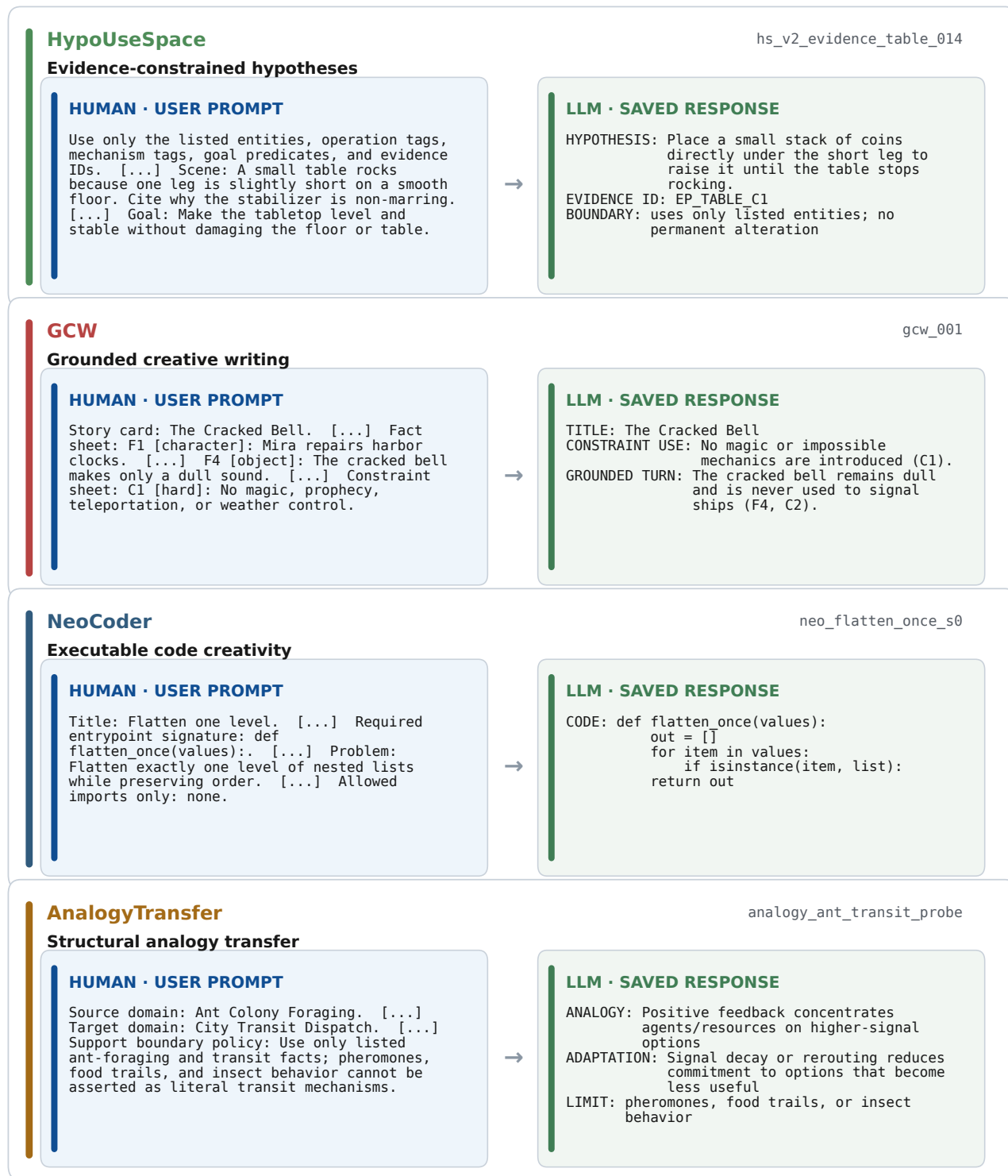


Figure N2: Human-LLM examples for HypoUseSpace, GCW, NeoCoder, and AnalogyTransfer. The left column reproduces passages from the evaluated user prompts; the right column reports selected field values from the saved responses. Bracketed ellipses mark omitted prompt text. Task IDs identify the full saved records, and colors carry no score information.

O Limitations

Correlation, Not Causation. Triple controls and per-item replication identify co-variation, not directional generative mechanism. A causal account would require interventional fine-tuning that selectively reduces one subtype, which we leave to follow-up work. The same controls reduce but cannot fully eliminate formula-driven correlation, since some atoms straddle the I/H boundary; cells whose estimate shifts across views are flagged in Appendix I.

The Machine–Human Coupling Gap. The two axes track human ratings individually, but the human ratings do not independently reproduce the negative coupling between them (-0.14 , 95% CI $[-0.37, +0.11]$, against -0.41 on the machine side, both reported in the main paper). A single 0–4 rating per axis may simply lack the resolution of sixteen subtype channels, and the human interval does not exclude a moderate negative coupling, but on present evidence the coupling is a property of WHITEBOARD’s scores that human raters have not confirmed at strength.

Measurement Resolution. Items carry 74.5% of the score variance and models 0.41%, giving $G=0.567$ and $\Phi=0.248$ at 80 items and leaving only 3.3% of model pairs separable. Absolute scores are therefore not comparable across item sets, and individual model rankings should not be read off these numbers. Five of the sixteen subtype channels also place more than half their outputs at the floor, which compresses the correlations those channels can express.

Scope and Resource Dependence. The map is only as general as the panel; generalization beyond 2026-vintage instruction-tuned LLMs is unclaimed. WHITEBOARD is deterministic and auditable but not resource-free: it depends on SBERT-class embeddings, association and feature norms, WordNet, curated common-answer banks, closed-world manifests, and hidden tests, so we name the scoring *deterministic and embedding-anchored* rather than symbolic. Proxy swaps of the embedding model and of bank coverage, run on the earlier prompt-collection panel, retained only a minority of decisive cells, so these resources carry structural weight on the

imagination side, and extension to another language requires rebuilding them rather than translating prompts. Deterministic atoms are also blind to higher-order qualities such as narrative tension, which a rubric-based judge would see at the cost of auditability.

P Statistical Power and Design Sensitivity

What effect sizes can 79 models and 80 shared items actually resolve? Because data collection was complete, we estimated design sensitivity rather than achieved power, in 10,000 Monte Carlo runs (seed 20260729), applying Spearman tests to Gaussian-copula samples of 79 models and paired Wilcoxon tests to variance-calibrated Gaussian random-effects samples for 58 complete models and 80 shared items. At nominal $\alpha=.05$, 80% power was reached at $|\rho|=.31$ for one coupling and an imagination-score gap of .041 ($d_z=.33$); conservative familywise references raised the thresholds to $|\rho|=.45$ for 63 cells and .078 ($d_z=.62$) for 1,653 pairs (Figure P1a–b). The selected cells have median raw $|\rho|=.56$, above the conservative correlation threshold, and shared-item pairing reduced the detectable imagination gap by 41% relative to the .069 unpaired threshold, a gain consistent with items carrying 74.5% of the score variance. The current design therefore has adequate sensitivity for the moderate-to-large couplings that support the main finding, while the shared anchor set improves model contrasts, although small controlled associations and closely spaced model pairs remain below its resolution.

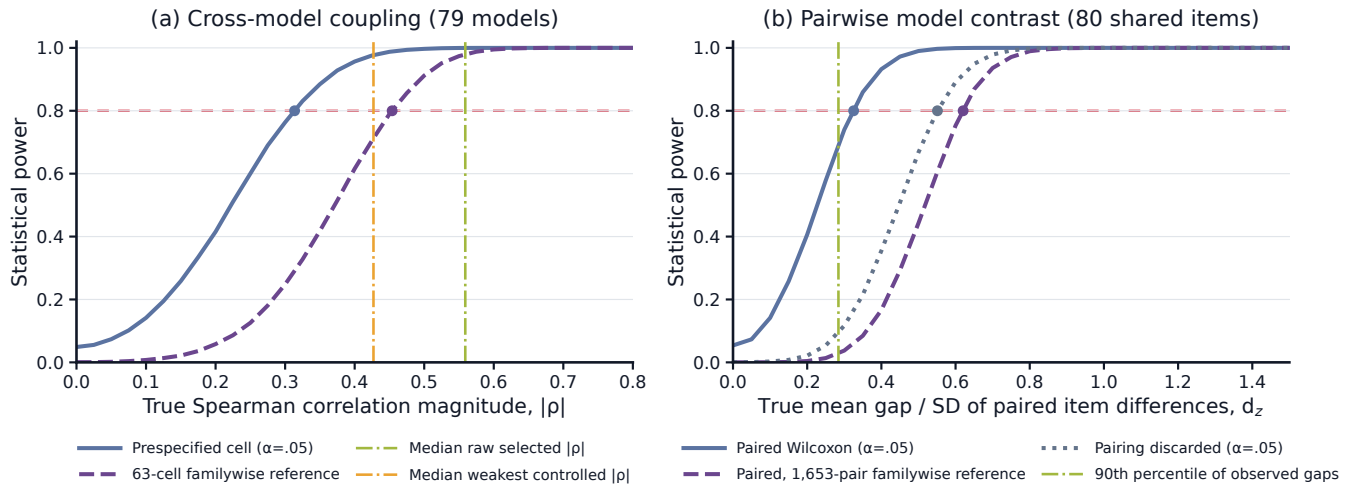


Figure P1: Statistical power under the current benchmark design. Panel (a) reports Monte Carlo power for a two-sided Spearman correlation over 79 models; the dashed curve uses $\alpha=.05/63$ as a conservative familywise reference for one 63-cell view, and the vertical lines mark the median raw and weakest controlled coefficients among the selected cells. Panel (b) reports variance-calibrated power for an imagination-score difference over 80 shared items; the curves compare the paired Wilcoxon test at $\alpha=.05$, a conservative $\alpha=.05/1653$ reference, and the Mann–Whitney U test after discarding item pairing. Points mark the minimum effect reaching 80% power, and the vertical line marks the 90th percentile of observed model-mean gaps.

Q Supplementary Related Work

The main paper focuses on work that measures imagination and hallucination together. This section expands that discussion by reviewing the two single-axis benchmark traditions that motivate WHITEBOARD’s paired design.

Q.1 Creativity Benchmarks for LLMs

Creativity benchmarks operationalize divergent production through several task-specific measures. The Divergent Association Task scores semantic distance among words generated to be mutually unrelated (Olson et al. 2021). Automated scoring of the Alternative Uses Test instead measures semantic distance between an alternative use and the object named in the prompt (Beatty et al. 2022). Forward flow measures how far each word in a free-association chain moves from earlier words (Gray et al. 2019). Stevenson et al. (2022) apply the Alternative Uses Test to GPT-3 and compare its outputs with human responses using expert ratings of originality, usefulness, surprise, and flexibility, together with automated semantic-distance scores. CreativityPrism groups tasks from divergent thinking, creative writing, and logical reasoning under quality, novelty, and diversity, using automatic judges validated against human annotations (Hou et al. 2026). LiveIdeaBench elicits scientific ideas from single-keyword prompts and evaluates them with a dynamic multi-model judging panel (Ruan et al. 2026). MacGyver tests object reuse under physical constraints (Tian et al. 2024), while CS4 varies the number of story-writing constraints to assess creativity without human ratings (Atmakuru et al. 2024). NeoCoder uses denial prompting to test divergent and convergent thinking on programming problems (Lu et al. 2025). Al Rabayah et al. (2025) test whether four LLM judges agree with an oracle set of Alternative Uses Test responses and with one another.

These benchmarks measure creativity or its evaluation, but they do not assign paired imagination and support-boundary labels to the same generation. WHITEBOARD adds that paired reading while retaining task-specific measures of divergent production.

Q.2 Hallucination Benchmarks for LLMs

Hallucination benchmarks range from answer-level truthfulness tests to fine-grained support checks. TruthfulQA tests whether models reproduce common human misconceptions in answers to adversarially selected questions, using generation and multiple-choice scores (Lin, Hilton, and Evans 2022). HaluEval provides generated and human-annotated hallucination examples for question answering, dialogue, and summarization (Li et al. 2023). FActScore decomposes long-form generations into atomic facts and verifies each against a knowledge source (Min et al. 2023). LongFact supplies open-domain prompts, while SAFE decomposes responses and checks individual facts with search (Wei et al. 2024). ALCE evaluates retrieval-augmented answers using citation correctness and completeness, alongside fluency and correctness (Gao et al. 2023). RAGTruth annotates hallucinations at both response and word levels in retrieval-grounded question answering and summarization (Niu et al. 2024). HALoGEN verifies atomic units across nine domains and distinguishes errors associated with incorrect recall, source knowledge, and fabrication (Ravichander et al. 2025). HalluLens separates intrinsic and extrinsic hallucination tasks (Bang et al. 2025), while Huang et al. (2025) organize definitions, causes, detection, and mitigation methods. Across these benchmarks, the unit of evaluation becomes progressively finer, but the labels remain on the truthfulness or support side. The cited benchmarks do not also assign an imagination subtype to

the same output. WHITEBOARD pairs that support-boundary reading with ten hallucination subtypes and seven imagination subtypes on each generation.

R Reliability and the Boundaries of the Instrument

We close with the resolution of the instrument itself. Among the 58 complete models, items account for 74.5% of variance and models for 0.41%; reliability is $G=0.567$ and $\Phi=0.248$. Five subtypes are compressed at the floor, while provider deletion preserves at least 70% of decisive cells in the gated and residual views (Appendix G). With only 3.3% of model pairs separable (section “The Anchor Set and Paired Discriminative Power” of the main paper), the instrument supports panel-level structure rather than individual rankings. Further boundaries are detailed in Appendix O.

References

- Al Rabeyah, A.; Góes, F.; Volpe, M.; and Medeiros, T. 2025. Do LLMs Agree on the Creativity Evaluation of Alternative Uses? In *Proceedings of the 16th International Conference on Computational Creativity*, 217–227.
- Anthropic. 2026. Model System Cards. Anthropic model documentation.
- Atmakuru, A.; Nainani, J.; Bheemreddy, R. S. R.; Lakkaraju, A.; Yao, Z.; Zamani, H.; and Chang, H.-S. 2024. CS4: Measuring the Creativity of Large Language Models Automatically by Controlling the Number of Story-Writing Constraints. arXiv:2410.04197.
- Bang, Y.; Ji, Z.; Schelten, A.; Hartshorn, A.; Fowler, T.; Zhang, C.; Cancedda, N.; and Fung, P. 2025. HalluLens: LLM Hallucination Benchmark. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 24128–24156.
- Beaty, R. E.; Johnson, D. R.; Zeitlein, D. C.; and Forthmann, B. 2022. Semantic Distance and the Alternate Uses Task: Recommendations for Reliable Automated Assessment of Originality. *Creativity Research Journal*, 34(3): 245–260.
- BenchLM. 2026. Seed 1.6 Model Profile. BenchLM model page, <https://benchlm.ai/models/seed-1-6>. Third-party profile; its current page reports no sourced benchmark row and is not evidence of an official LMArena score.
- Buchanan, E. M.; Valentine, K. D.; and Maxwell, N. P. 2019. English semantic feature production norms: An extended database of 4,436 concepts. *Behavior Research Methods*, 51(4): 1849–1863. Data: <https://osf.io/cjyzw/>; reference code: <https://github.com/doomlab/Word-Norms-2>.
- ByteDance Seed. 2025. Seed1.5-Thinking: Advancing Superb Reasoning Models with Reinforcement Learning. arXiv:2504.13914.
- ByteDance Seed. 2026. Seed2.0. Official model page and model card.
- Chiang, W.-L.; Zheng, L.; Sheng, Y.; Angelopoulos, A. N.; Li, T.; Li, D.; Zhu, B.; Zhang, H.; Jordan, M.; Gonzalez, J. E.; and Stoica, I. 2024. Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 8359–8388. PMLR.
- De Deyne, S.; Navarro, D. J.; Perfors, A.; Brysbaert, M.; and Storms, G. 2019. The “Small World of Words” English word association norms for over 12,000 cue words. *Behavior Research Methods*, 51(3): 987–1006. SWOW-EN2018 data access: <https://smallworldofwords.org/en/project/research>.
- DeepSeek. 2026. Change Log. DeepSeek API documentation.
- Gao, T.; Yen, H.; Yu, J.; and Chen, D. 2023. Enabling Large Language Models to Generate Text with Citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 6465–6488.
- GLM Team. 2025. GLM-4.7, GLM-4.6, and GLM-4.5. Official model repository.
- Google DeepMind. 2026. Model Cards. Google DeepMind model documentation.
- Gray, K.; Anderson, S.; Chen, E. E.; Kelly, J. M.; Christian, M. S.; Patrick, J.; Huang, L.; Kenett, Y. N.; and Lewis, K. 2019. “Forward Flow”: A New Measure to Quantify Free Thought and Predict Creativity. *American Psychologist*, 74(5): 539–554.
- Hou, Z. J.; Zhang, B. A.; Lu, Y.; Baghel, B. K.; Brei, A.; Lu, X.; Jiang, M.; Brahman, F.; Chaturvedi, S.; Chang, H.-S.; Khashabi, D.; and Li, X. L. 2026. CreativityPrism: A Cross-Domain Evaluation Framework for Large Language Model Creativity. *Transactions on Machine Learning Research*.
- Huang, L.; Yu, W.; Ma, W.; Zhong, W.; Feng, Z.; Wang, H.; Chen, Q.; Peng, W.; Feng, X.; Qin, B.; and Liu, T. 2025. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. *ACM Transactions on Information Systems*, 43(2): 1–55.
- Li, J.; Cheng, X.; Zhao, X.; Nie, J.-Y.; and Wen, J.-R. 2023. HaluEval: A Large-Scale Hallucination Evaluation Benchmark for Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 6449–6464.
- Lin, S.; Hilton, J.; and Evans, O. 2022. TruthfulQA: Measuring How Models Mimic Human Falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 3214–3252.
- LM Arena. 2026. Arena Leaderboard Dataset. Hugging Face Datasets, <https://huggingface.co/datasets/lmarena-ai/leaderboard-dataset>. Text-style-control snapshot published 2026-04-27; revision 6df7986d4be040bf3a736238cc437c17fb259671; CC BY 4.0; accessed 2026-04-29.
- Lu, Y.; Wang, D.; Li, T.; Jiang, D.; Khudanpur, S.; Jiang, M.; and Khashabi, D. 2025. Benchmarking Language Model Creativity: A Case Study on Code Generation. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2776–2794.
- Meta. 2024. Llama 3.3 Model Card. Official model card.

- Miller, G. A. 1995. WordNet: A Lexical Database for English. *Communications of the ACM*, 38(11): 39–41. WordNet 3.0 license: <https://wordnet.princeton.edu/license-and-commercial-use>.
- Min, S.; Krishna, K.; Lyu, X.; Lewis, M.; Yih, W.-t.; Koh, P.; Iyyer, M.; Zettlemoyer, L.; and Hajishirzi, H. 2023. FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 12076–12100.
- MiniMax. 2026a. MiniMax M2.5: Built for Real-World Productivity. Official release blog.
- MiniMax. 2026b. MiniMax M3: Frontier Coding, 1M Context, Native Multimodality—All in One Model. Official release blog.
- Moonshot AI. 2026a. Kimi K2.5. Official model repository and technical report.
- Moonshot AI. 2026b. Kimi K2.6: From Code to Creation, From One to Many. Official model page.
- Niu, C.; Wu, Y.; Zhu, J.; Xu, S.; Shum, K.; Zhong, R.; Song, J.; and Zhang, T. 2024. RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 10862–10878.
- Olson, J. A.; Nahas, J.; Chmoulevitch, D.; Cropper, S. J.; and Webb, M. E. 2021. Naming unrelated words predicts creativity. *Proceedings of the National Academy of Sciences*, 118(25): e2022340118.
- OpenAI. 2026. All Models. OpenAI API documentation.
- Qwen Team. 2026a. Qwen3.5: Towards Native Multimodal Agents. Official release blog.
- Qwen Team. 2026b. Qwen3.6-35B-A3B: Agentic Coding Power, Now Open to All. Official release blog.
- Ravichander, A.; Ghela, S.; Wadden, D.; and Choi, Y. 2025. HALoGEN: Fantastic LLM Hallucinations and Where to Find Them. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 1402–1425.
- Reimers, N.; and Gurevych, I. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, 3982–3992.
- Ruan, K.; Wang, X.; Hong, J.; Wang, P.; Liu, Y.; and Sun, H. 2026. Evaluating LLMs’ Divergent Thinking Capabilities for Scientific Idea Generation with Minimal Context. *Nature Communications*, 17(1): 3625.
- Sentence Transformers. 2025. all-mpnet-base-v2 Model Card. Hugging Face model repository, <https://huggingface.co/sentence-transformers/all-mpnet-base-v2/tree/e8c3b32edf5434bc2275fc9bab85f82640a19130>. Revision e8c3b32edf5434bc2275fc9bab85f82640a19130; Apache License 2.0; accessed 2026-04-28.
- Stevenson, C.; Smal, I.; Baas, M.; Grasman, R.; and van der Maas, H. 2022. Putting GPT-3’s Creativity to the (Alternative Uses) Test. In *Proceedings of the 13th International Conference on Computational Creativity*, 164–168.
- Tencent Hy Team. 2026. Hy3. Official model repository.
- Tian, Y.; Ravichander, A.; Qin, L.; Le Bras, R.; Marjeh, R.; Peng, N.; Choi, Y.; Griffiths, T.; and Brahman, F. 2024. MacGyver: Are Large Language Models Creative Problem Solvers? In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 5303–5324.
- Wei, J.; Yang, C.; Song, X.; Lu, Y.; Hu, N.; Huang, J.; Tran, D.; Peng, D.; Liu, R.; Huang, D.; Du, C.; and Le, Q. V. 2024. Long-form Factuality in Large Language Models. In *Advances in Neural Information Processing Systems*, volume 37, 80756–80827.
- xAI. 2026. Models. xAI developer documentation.
- Xiaomi MiMo. 2026. MiMo-V2.5 Series Release Notice. Official API platform.
- Z.ai. 2026. GLM-5: From Vibe Coding to Agentic Engineering. Official release blog.